MOHAMMAD KAMRUL HASAN

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Education

Bangladesh University of Engineering and Technology (BUET) Apr 2025 – 2028 (expected)
B.Sc. in Computer Science and Engineering • CGPA 3.77/4.00 *Dhaka, Bangladesh*
Ranked 79th in the national BUET admission merit list

Experience

Backend Developer Intern Jun 2026 – Present
Version3.ai • *conversational-commerce AI for online stores* *Dhaka, Bangladesh*

- Built the multi-tenant product database service (**NestJS**, **MongoDB**, **RabbitMQ**) with per-store provisioning and strict data isolation, so no store can ever read another store's catalogue.
- Designed the **WooCommerce** and **Shopify** integrations so a store connects in one click: **OAuth 2.0 PKCE**, HMAC-signed requests, AES-256-GCM secret storage, a durable dispatch queue in place of WP-Cron, and idempotent webhooks so retries are never processed twice.
- Stopped WooCommerce from permanently deleting every connecting store's API keys by tracing it to the OAuth callback returning `201` instead of `200`.
- Chose the visual product-search model by benchmarking **4** candidates (SigLIP2, Marqo, fashion-tuned CLIP) on Products-10K (**9,691** SKUs, **132K** images) with Recall@k, MRR, nDCG@10 and bootstrap CIs; the winner reached **Recall@1 0.672**.
- Proved search can run on CPU-only servers: **Qdrant** int8/binary quantization cost a median **0.004** Recall@1, and **4** single-thread workers gave **2.2**× the throughput of one 4-thread worker.

Software Engineering Intern Dec 2025 – Mar 2026
Edulytics *Dhaka, Bangladesh*

- Completed the mobile app MVP in **React Native** and **TypeScript** from the team's designs, and built features for the web app, shipping through feature branches and PR review with a remote team.

Projects

LITTLETai — core-aware thread presets for llama.cpp on Android 2026
Arm Create: AI Optimization Challenge (submitted) • POSIX shell, Python, simpleperf, Perfetto • github.com/Kamrulhasan12345/LITTLETai

- Made **llama.cpp** decode **2.2–2.6× faster** (8.06 → 17.9–20.8 tok/s) with **2.6–3.3× less energy** on a Helio G81 phone (Qwen2 1B Q4_0), with a root-free shell tool that groups cores by frequency and MIDR and emits thread and affinity presets.
- Found why the default was slow with **simpleperf**: instructions per token rose from ∼143M at 2 threads to 365–705M at 8 as threads spin-waited in `ggml_barrier()`; llama.cpp's affinity flag was also silently ignored on Android (99.5% of cycles stayed on the big cores), so I pinned with `taskset`.
- Made every number reproducible: a seeded, randomized harness with thermal gates, energy logging and per-core PMU counters regenerates all **238** configurations byte-for-byte, backed by **209** tests.

Ilham — hadith study platform 2026
Database course project, team of 2 • PostgreSQL, PL/pgSQL, Node.js, Express, TypeScript • github.com/Kamrulhasan12345/ilham

- Loaded **14,941** hadiths, **139,766** chain positions and **20,957** narrators with **99.6%** narrator resolution, via an ETL into a three-schema **PostgreSQL** design (staging, read-only corpus, study layer).
- Scored each hadith's reliability in SQL from the weakest narrator in its chain, and locked the corpus with `REVOKE` so the app can never corrupt reference data; covered by a **13**-check smoke test.

Sentinel — offline medical triage app for Android 2025
Lead developer, team of 3 • React Native, TypeScript, TensorFlow Lite • github.com/Kamrulhasan12345/sentinel

- Triages symptom descriptions into **52** first-aid intents and Critical/Urgent/Routine **fully offline**, so it works with no signal: an on-device **TF Lite** 1D-CNN classifies first, falling back to int8 MiniLM embeddings (**20 MB**) below **65%** confidence. Released as an APK.

Bayyinah — collaborative desktop Quran app 2026
Java, JavaFX, Spring Boot, WebRTC, SQLite FTS5, PostgreSQL • github.com/Kamrulhasan12345/bayyinah

- Built offline full-text search over all **6,236** verses (**SQLite FTS5**) in a JavaFX client, so it works with no connection, plus live group reading with **WebRTC** audio and STOMP signaling on **Spring Boot**; 190+ commits.

Publication

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka
UbiComp/ISWC 2026, ACM • co-author, BUET × UC San Diego • I transcribed the interviews behind the qualitative analysis • doi.org/10.1145/3830727.3834827

Skills

Languages: C, C++, Python, TypeScript, JavaScript, Java, SQL (PostgreSQL, PL/pgSQL), POSIX shell
ML & inference: llama.cpp, GGUF quantization, PyTorch, TensorFlow Lite, Hugging Face Transformers, sentence-transformers, SigLIP/CLIP, FAISS, Qdrant
Backend & data: NestJS, FastAPI, Spring Boot, Express, REST, WebSockets, WebRTC, OAuth 2.0, JWT, MongoDB, Redis, RabbitMQ, SQLite
Tools: Linux, Docker, Git, GitHub Actions, simpleperf, Perfetto, adb